Henry Bao

+1-437-345-2716 | h39bao@uwaterloo.ca | https://www.linkedin.com/in/henryhbao | https://github.com/TruVortex

Education

University of Waterloo Waterloo, ON
Bachelor of Computer Science *Sept. 2025 – Apr. 2030*

Technical Skills

Languages: C/C++, Java, Python, JavaScript/Typescript, HTML/CSS, SQL, Golang, Rust, OCaml, Bash
Frameworks: Next.js, React, Node.js, Tailwind, Django, Flask, MediaPipe, Solana, FastAPI, RESTful APIs
Developer Tools: Git, VS Code, Intellij IDEA, CMake, Linux, Docker, Claude Code, Cursor, Terraform, Azure, Excel
Libraries: TensorFlow, Keras, PyTorch, scikit-learn, JAX, pandas, NumPy, Matplotlib, ffmpeg, PostgreSQL

Experience

Infrastructure Support Specialist May 2026 – Now
Teranet *Toronto, ON*

- Acted as the primary responder for alerts and incidents from Spectrum, Solarwinds, CommVault, etc.
- Executed time-sensitive patching tasks during planned maintenance windows with zero unexpected downtime
- Fulfilled change requests for server builds and decommissions across various hardware, host (on-prem and Azure), and operating system configurations, allowing for system provisioning and hardening
- Streamlined deployments and automated tasks using the infrastructure as code (IaC) tools Terraform and Ansible
- Performed ad-hoc tasks such as optimizing Elasticsearch for the application team using ES|QL

Projects

ConstexprJIT | *C++, CMake* Jul. 2026

- Built a C++23 compile-time JIT that emits executable x86-64 machine code from expressions using consteval
- Designed a constexpr pipeline and SSE2 backend to generate machine code without dynamic memory
- Wrote a runtime loader to manage relocation patching and dynamic symbol resolution for external function calls
- Implemented a memory manager with W^X-compliant protection to safely allocate and execute compiled code

Truv Compression | *Rust* Jun. 2026

- Developed a compression tool based on the DEFLATE standard, utilizing LZ77 and dynamic Huffman coding
- Enabled multi-threaded processing with rayon, partitioning files into independent blocks for faster compression
- Accelerated match finding by replacing byte-by-byte comparisons with 64-bit word operations through SWAR

Tiny Compiler | *OCaml, Dune* Jun. 2026

- Built a functional language compiler supporting lexical scoping, first-class functions, conditionals, and let-bindings
- Implemented Hindley–Milner type inference (Algorithm W) with unification, polymorphic let-generalization, instantiation, and occurs-check detection for fully inferred static typing
- Wrote optimization passes for beta reduction, closure conversion, constant folding, and dead branch removal
- Created a stack-based virtual machine and bytecode engine to manage runtime execution and environment capture

Tiny Binary Serializer | *OCaml, Dune, Bigstringaf* May 2026

- Developed a binary serializer using GADTs to enforce data schema safety at compile-time
- Streamlined memory usage by supporting zero-copy buffer slices and data structures like product and sum types
- Reduced the storage footprint of metadata by implementing variable-length integer encoding

p2p-file-sync | *Golang, Claude Code* Feb. 2026

- Developed a decentralized file sync engine using Merkle Trees to reconcile state across peers in $O(\log n)$ time
- Reduced bandwidth and memory usage by implementing content-addressable storage with SHA-256 chunking
- Implemented a delta sync protocol using rolling hashes to isolate and transfer only the modified regions of a file
- Encrypted peer-to-peer networking over QUIC, using STUN-based UDP hole punching for NAT traversal

Achievements

CEMC Contests: 14 honour rolls in mathematics and computing contests including one first-place and one runner-up
MAA Contests: 3 time qualification to the AIME mathematics contest
Advanced Coursework: 2x 4.0 and 4x 3.9 in advanced math and computer science courses
Certifications: Microsoft Certified: Azure Fundamentals (AZ-900) and HashiCorp Certified: Terraform Associate (004)